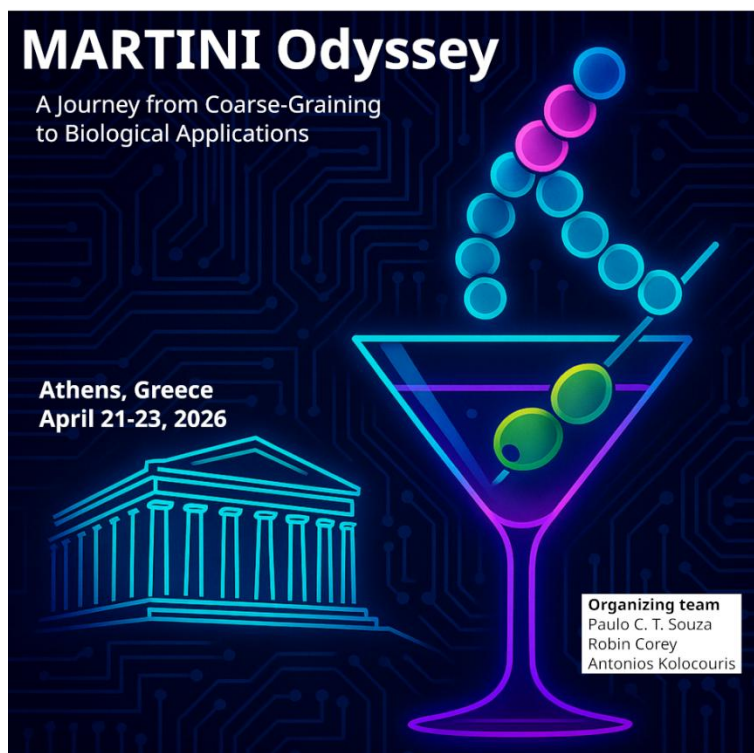


BOOK OF POSTER ABSTRACTS & SCIENTIFIC PROGRAM



Erasmus+

ELPEN

P1: MARTINI-based Modeling of Peptide-Membrane Interactions for Advanced Tissue Engineering Applications

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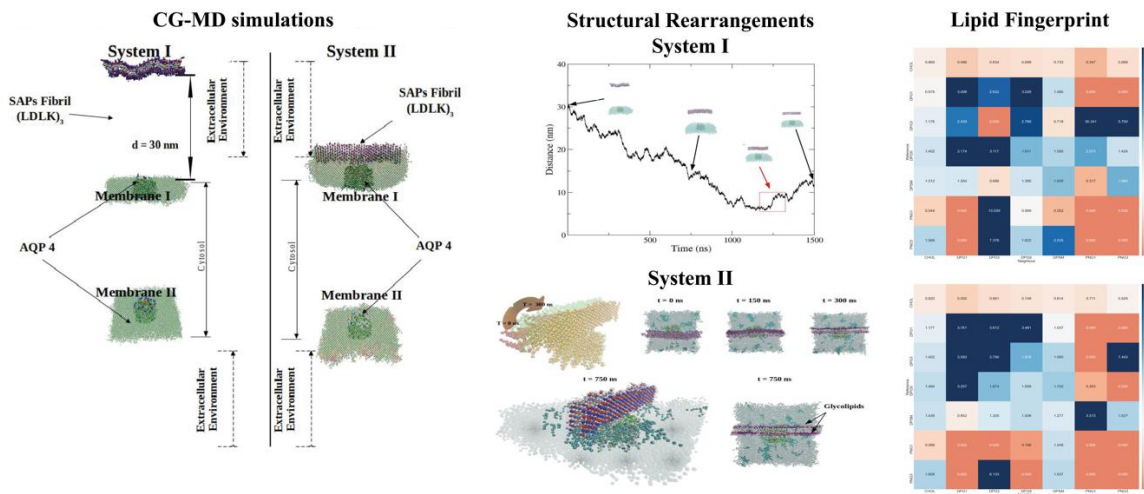
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Abstract

Self-assembling peptide (SAP) hydrogels have shown great potential in treating spinal cord and brain injuries, particularly when combined with adult neural stem cells. These hydrogels provide a biocompatible and tunable environment that supports cell survival, proliferation, and integration into host tissues. To advance their application in tissue engineering and regenerative medicine, it is essential to understand how mechanochemical signaling influences cellular differentiation, particularly in response to the hydrogel matrix's dynamic physical and chemical cues.

A critical aspect of this involves elucidating molecular interactions at the neuronal membrane, which remains a promising area of research for both drug delivery and biomaterial design. Computational simulations of peptide–membrane interactions provide a powerful means to investigate these phenomena at the molecular level. [1] By capturing the dynamics, insertion, and orientation of peptides within complex lipid bilayers, molecular dynamics (MD) and MARTINI coarse-grained modeling approaches shed light on the mechanisms underlying membrane disruption, signaling modulation, and selective targeting. [2][3]

These techniques allow exploration of time and length scales inaccessible to experimental methods, offering detailed insight into the physicochemical determinants of peptide functionality. An innovative MARTINI-based MD framework explored how SAP fibrils with varying surface charges affect the organization and dynamics of lipid domains in neural membranes. [2] This approach enables the *in-silico* characterization of the biomimetic properties of SAP hydrogels and related polymeric materials, facilitating their rational optimization for specific neurological applications. [2][3] Together, these advances position computational modeling as a central tool in developing next-generation therapeutics and synthetic biointerfaces for tissue engineering.



Keywords: Self-Assembling Peptides, Coarse-Grained MD, Neuronal Membranes, MARTINI force-field

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P2: Integrating All-Atom and Coarse-Grained Simulations to Decipher the Interactions of the MP1 Peptide with Model Membranes

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Abstract

Bioactive peptides play a crucial role in various biological processes, including host defense against microorganisms. Their mechanism of action typically involves electrostatic and hydrophobic interactions with cell membranes, perturbing lipid packing and inducing the formation of pores or defects that lead to leakage or cell death. An example of such molecules is the Polybia-MP1 (MP1), which exhibits broad-spectrum bactericidal activity against both Gram-positive and Gram-negative bacteria, while remaining non-hemolytic and non-cytotoxic to eukaryotic cells. The MP1 peptide stands out for its selective action against bacteria and tumor cells, targeting anionic phospholipids like phosphatidylglycerol (PG) and phosphatidylserine (PS). Although its macroscopic effects are known, the molecular details of its collective mechanism, specifically regarding pore formation and the impact of membrane complexity, remain to be fully elucidated. In this work, we employ a multiscale computational strategy to investigate the collective behavior of MP1 and its analogues.

Preliminary All-Atom (AA) molecular dynamics simulations characterize the initial adsorption of individual peptides, identifying local perturbations such as changes in lipid order parameters and bilayer curvature. To overcome the temporal and spatial limitations of the atomic scale, we transition to the Coarse-Grained (CG) framework. This approach allows us to simulate systems with multiple peptides to observe concentration-dependent events, including peptide aggregation and the possible formation of transmembrane pores. In the CG approach, we first benchmark the model by investigating the interactions of MP1 with simple lipid bilayer compositions, such as pure phosphatidylcholine (PC) and phosphatidylethanolamine (PE) bilayers. These systems were selected because experimental evidence is available regarding MP1–membrane interactions, including subtle shifts in the phase transition temperature (T_m) of PE bilayers. Following this initial validation, we plan to extend our simulations to more complex membrane compositions that better mimic bacterial and cancer cell membranes. By integrating AA and CG perspectives, including backmapping strategies, we aim to identify the structural features that optimize lytic action and selectivity. This study seeks to provide a molecular basis for the design of more potent therapeutic analogues, bridging the gap between specific atomic interactions and large-scale membrane processes.

This work was supported by CNPq (404205/2024-0, 174388/2024-1, and 201531/2025-9). Computational resources were provided by the National Laboratory for Scientific Computing (LNCC) through the Santos Dumont supercomputer. We also acknowledge the support of the PSMN (Pôle Scientifique de Modélisation Numérique) and the Centre Blaise Pascal's IT test platform at ENS de Lyon (Lyon, France) for the computer facilities. The platform operates the SIDUS solution developed by Emmanuel Quemener.

P3: Bridging Simulation Scales to Map the ACE2 Conformational Landscape

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Abstract

Angiotensin-Converting Enzyme 2 (ACE2) is a membrane-embedded protein that is the primary receptor for SARS-CoV-2. Experimental evidence indicates that the membrane lipid composition, particularly cholesterol enrichment, affects viral entry into host cells. However, the molecular mechanisms by which the membrane phase modulates receptor behaviour remain insufficiently characterised. In this work, we first leveraged multi-replica all-atom molecular dynamics (MD) simulations to demonstrate that the membrane composition reshapes the ACE2 conformational free-energy landscape. Three different membrane compositions were employed: fluid (phospholipids), intermediate (80% phospholipids, 20% cholesterol), and raft-like membranes (30% phospholipids, 50% cholesterol, and 20% sphingolipids). We show that phospholipids-only fluid bilayers promote high extracellular domain (ECD) flexibility and bent orientations, which may occlude the spike-binding epitope and hinder viral attachment. In contrast, cholesterol-rich raft-like environments and the ACE2 dimeric state restrict these motions, favouring a geometry that supports spike engagement. To extend these findings to longer timescales, we employed the GōMartini3 coarse-grained (CG) framework with optimised contact maps based on high-frequency contacts (HFC) (Olivos-Ramirez, Poma et al., 2026, DOI: 10.1101/2025.11.14.688437), which were obtained from all-atom trajectories. Using modified contact maps significantly enhances the sampling of conformational space compared to the control model with a standard contact map derived from a single-structure model. This multiscale approach provides a quantitative link between the lipid membrane composition and ACE2 receptor conformational landscape, providing key insights into the molecular mechanisms that govern receptor dynamics.

Acknowledgements

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P4: Assessing the Performance of Non-Equilibrium Thermodynamic Integration in Flavodoxin Redox Potential Estimation

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Abstract

Flavodoxins are enzymes that contain the redox-active flavin mononucleotide (FMN) cofactor and play a crucial role in numerous biological processes, including energy conversion and electron transfer. Since the redox characteristics of flavodoxins are significantly impacted by the molecular environment of the FMN cofactor, the evaluation of the interplay between the redox properties of the flavin cofactor and its molecular surroundings in flavoproteins is a critical area of investigation for both fundamental research and technological advancements, as the electrochemical tuning of flavoproteins is necessary for optimal interaction with redox acceptor or donor molecules. In order to facilitate the rational design of biomolecular devices, it is imperative to have access to computational tools that can accurately predict the redox potential of both natural and artificial flavoproteins. In this study, we have investigated the feasibility of using non-equilibrium thermodynamic integration protocols to reliably predict the redox potential of flavodoxins. Using as a test set the wild-type flavodoxin from *Clostridium beijerinckii* and eight experimentally characterized single-point mutants, we have computed their redox potential. Our results show that 75% (6 out of 8) of the calculated reaction free energies are within 1 kcal/mol of the experimental values, and none exceed an error of 2 kcal/mol, confirming that non-equilibrium thermodynamic integration is a trustworthy tool for the quantitative estimation of the redox potential of this biologically and technologically significant class of enzymes.

P5: Drug Repurposing; In silico Investigation of Sartans as Potential Candidates for the Treatment of Alzheimer's disease

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Abstract

Alzheimer's disease is a multifactorial ailment that affects millions of people worldwide. Multiple theories have emerged to shed light on the disease's pathophysiology. Thus, the need for multi-target-directed ligands to combat this condition has emerged. In Alzheimer's disease (AD) brains, the expression of MAO-B is increased in the hippocampus and cerebral cortex compared to healthy brains, while its activation plays a role in the formation of amyloid- β (A β) plaques, production of free radicals and hydrogen peroxide, leading to oxidative damage of neurons. For years, terminal propargylamines have been used against monoamine oxidase-B. Our research team repurposed well-known angiotensin II receptor blockers, sartans, and tested them against the MAO-B enzyme. This was achieved through *in silico* docking methods. Moreover, their mode of binding was compared to that of known terminal propargylamines as well as the known drug safinamide. Sartans showcased exceptional affinity for the MAO-B enzyme. *In vitro* experiments are on the way to prove these computational results.

P6: Study of the GHRHR Activation Mechanism via Molecular Dynamics Simulations

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Abstract

The Growth Hormone Releasing Hormone Receptor (GHRHR) is a member of the class B1 G-protein-coupled receptor (GPCR) subfamily.¹ GHRHR is activated by its cognate 44-amino-acid peptide ligand, GHRH. GHRHR signaling has clinically relevant links to oncogenesis: GHRH/GHRHR and tumor-expressed splice variants (e.g., SV1) form local autocrine/paracrine growth circuits in a wide spectrum of human cancers, and GHRH antagonists have demonstrated antiproliferative effects in many experimental cancer models.² GHRHR consists of a large N-terminal extracellular domain (ECD) that confers ligand specificity, a canonical seven-transmembrane (7TM) domain that mediates signal transduction, and an intracellular C-terminal (CT) tail. Upon ligand binding, the N-terminal portion of GHRH engages the ECD, followed by insertion into the 7TM core, initiating receptor activation³ which involves hallmark rearrangements of the transmembrane bundle, including outward movement of TM6 to open the intracellular G-protein-binding site and complementary adjustments of TM1 and TM7 to stabilize the active state.⁴

Although structural information obtained to date has advanced our understanding of GHRHR, the detailed mechanism by which ECD engagement is transmitted through the 7TM domain to the intracellular G-protein interface remains unknown. Here, we employ molecular dynamics simulations of full-length GHRHR to gain atomistic insights of the activation mechanism of GHRHR. Our analyses indicate that distal domains, including the CT tail and ECD, contribute to receptor activation by forming transient interactions with intracellular loops and the 7TM core. These dynamic contacts appear to modulate the accessibility of the G-protein-binding site and influence the conformational equilibria of the receptor. These findings provide new insights into the mechanistic interplay between distal regulatory elements and the core activation machinery of GHRHR and may guide the design of modulators of receptor activation.

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P7: Hybrid Palladium-Tryptophan-Solid Lipid Nanoparticles: Experimental and numerical study

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Abstract

Lipid-based nanoparticles are being extensively studied for biomedical applications, due to their high biocompatibility and potential for drug delivery. In this work, hybrid lipid-metal systems in the form of nanoparticles (NPs) are studied for use as antitumor agents. The hybrid NPs consist of Palladium nanoparticles functionalized with essential amino acid Tryptophan (PdTrp NPs) and integrated into solid lipid nanoparticles (SLNPs) based on stearic acid. The PdTrp-SLNPs system has been synthesized via the electrospray method and characterized with various spectroscopy and microscopy techniques. Interactions between the components of the hybrid system have been studied by metadynamics and molecular dynamics simulations. The PdTrp-SLNPs are stable with optimal fluorescent properties, having a higher retention of Tryptophan molecules compared to Trp-SLNPs without the metal component. Preliminary results from metadynamics simulations in different solvents based on water and ethanol mixtures indicated a change in preferred orientations and energies of adsorption of the Tryptophan molecule on Palladium surfaces. Further studies will focus on interactions of functionalized Pd-TRP NPs with lipids, with the goal to model the complete PdTrp-SLNP hybrid system.

P8: P2Y12 receptor goes toward activation: insights by Principal Component Analysis

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Abstract

The G-protein-coupled receptor P2Y12, prominently expressed on the cellular membrane of platelets, serves as the primary target for existing antiaggregant medications. Despite the established efficacy of these pharmaceuticals in clinical practice, critical concerns persist, notably concerning fatal bleeding events and variable effectiveness across patient populations. Furthermore, the molecular mechanisms underlying the actions of these drugs remain elusive. To improve our knowledge and for future drug design, this study employs a computational approach to investigate the molecular interactions between the full agonist 2MeSADP and the P2Y12 receptor at a molecular level, to elucidate the P2Y12R activation mechanism. Principal Component Analysis was used to pinpoint the Essential Dynamics of P2Y12, and the Gibbs free energy landscape was plotted. This procedure allowed us to identify the most visited energy minima and to uncover the P2Y12R mechanism towards activation in response to the full agonist 2MeSADP, through the identification of precise molecular switches, which cause the rearrangement of the helical bundle, in particular helices V and VI. By inspecting the movement along different eigenvectors, the essential dynamics show the movement of helix V from a bent to a vertical position, and the slight outward movement of helix VI. These two decomposed movements are crucial as they facilitate the creation of a crevice for G-protein accommodation. Finally, by comparing these findings with experimental structural data, this computational study seems to be aligned with the currently available literature about the P2Y12R, paving the way to the study of the endogenous agonist, ADP.

P9: What are the challenges of hybrid all-atom/coarse-grained protein simulations? Example of fluorescent proteins containing biliverdin

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Abstract

Multi-scale molecular simulations are especially useful for investigating large protein systems and capturing processes that occur over long timescales. In this work, a hybrid all-atom/coarse-grained force field was developed by combining the coarse-grained SIRAH¹ force field with the atomistic ff14SB² force field, with a specific focus on protein systems. The methodology previously established for DNA³ hybrid force field development was adapted and applied to protein systems. As a test case, we focus on bacteriophytochromes, fluorescent proteins that contain a covalently bound biliverdin chromophore responsible for light absorption and emission. Biliverdin can adopt two main conformations, referred to as extended and compact states, with the compact form exhibiting a red-shifted absorption spectrum. Understanding which amino acids stabilize each conformation is therefore key to guiding mutations toward more red-shifted variants. However, these conformational transitions occur on long timescales and typically require computationally expensive approaches such as umbrella sampling or adaptive steered molecular dynamics. In this context, hybrid all-atom/coarse-grained strategies offer an alternative by significantly reducing the computational cost while retaining essential molecular detail. The workflow included the construction of hybrid protein systems, mapping between atomistic and coarse-grained representations, and the setup of molecular dynamics simulations. Particular attention was given to the consistency between resolutions. Several challenges were encountered during this process, including the treatment of the interface between atomistic and coarse-grained regions, parameter compatibility, equilibration strategies, and the definition of appropriate validation metrics for hybrid systems. These difficulties highlight broader methodological questions in multi-scale modeling of proteins. Although this work was conducted using SIRAH force field, it raises important questions regarding how such approaches could be translated into the Martini 3 framework. In particular, we are interested in understanding how protein systems are best treated in Martini, how parametrization and system setup differ, and what strategies are recommended for ensuring stability and physical relevance in hybrid simulations. This poster presents the methodology used to construct the hybrid model, discusses the main challenges encountered, and outlines key open questions that motivate further exploration of hybrid approaches, particularly within the Martini system.

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P10: Angiotensin 1-7 Modulates the Dynamics and Activation of the Protooncogene Mas Receptor

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Abstract

The proto-oncogene Mas receptor (MasR) is a class-A G protein-coupled receptor (GPCR) that mediates the cardioprotective effects of Angiotensin 1-7 (Ang 1-7) within the renin-angiotensin system (RAS). Despite its therapeutic relevance, the molecular mechanisms underlying MasR activation remain poorly understood due to the lack of experimentally resolved structural data. In this study, we combined 1-microsecond all-atom molecular dynamics simulations of an AlphaFold-predicted MasR model with energetic network analyses to investigate ligand-induced activation dynamics. Our results challenge the conventional electrostatic clamp model and reveal a distinct “active anchor” mechanism. We demonstrate that Ang 1-7 releases the inactive core lock (ARG245–TYR248) to engage a critical anchor cluster (SER109/PHE112). In this state, the NPxxY motif becomes fully activated, and the intracellular vestibule expands through non-canonical communication hubs, while TM6 remains in a relatively compact conformation (~10 Å). Thermodynamic analysis indicates that this partially activated state is supported by favorable binding energetics and increased conformational plasticity. These findings provide a structural explanation for the partial agonist profile of Ang1-7 and propose a mechanistic framework for therapeutic targeting of the ACE2/Ang-(1-7)/MasR axis. We propose that effective drug design strategies need not aim for maximal receptor opening, but rather stabilization of this compact active state using peptidomimetics or positive allosteric modulators (PAMs).

Keywords: MasR, Angiotensin 1-7, GPCR, Receptor activation, Molecular modeling

P11: Molecular Dynamics Study of Mutation Effects on Human Soluble Epoxide Hydrolase

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Abstract

Epoxyeicosatrienoic acids (EETs) are bioactive lipid mediators with anti-inflammatory, vasodilatory, and analgesic properties. Their levels are regulated by soluble epoxide hydrolase (sEH), which converts EETs into less active dihydroxyeicosatrienoic acids. Given its involvement in inflammation, cardiovascular, and neurological disorders, sEH has emerged as a promising therapeutic target. In this study, we investigate the effect of four sEH polymorphisms (Lys55Arg, Arg103Cys, Cys154Tyr, Arg287Gln) on protein stability and inhibitor binding. For this purpose, all-atom molecular dynamics simulations were performed in the Schrödinger Maestro 14.3 suite. Following protein preparation and in silico mutagenesis, systems were solvated, neutralized, and simulated with Desmond (OPLS4 force field) for 500 ns under NPT conditions. The simulations did not reveal significant mutation-dependent alterations in inhibitor binding. This is consistent with the fact that the polymorphisms are located away from the active site, while key binding residues (D335, P361, Y383, N472, Y466) directly mediate inhibitor interactions. These results suggest that the current 500 ns timescale may not be sufficient to capture subtle or slow-evolving conformational changes induced by distal mutations. Future work will focus on applying coarse-grained approaches to reduce computational cost and enable extended simulations, potentially allowing the study of the dimeric state, which represents the biologically active form of the protein.

P12: AI-assisted unbiased docking as a tool for prediction of hERG channel block liability

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Abstract

Pharmacological inhibition of cardiac hERG potassium channels by diverse small molecule drugs is causally linked to acquired long QT syndrome and torsades de pointes arrhythmia. Accordingly, all new drugs must be evaluated for potential hERG inhibition. Pharmacophore-based computational prediction models of hERG block by drugs are not sufficiently accurate at this time to supplant time-consuming electrophysiological evaluation of hERG inhibition. Prediction models that are channel structure-based may improve accuracy and granularity with respect to computationally predicted drug-channel interactions. However, at present, structure-based models are constrained and ‘biased’ as they require a user-specified (preconceived) starting configuration (typically in the channel’s central cavity) when docking drug molecules, which may miss important interactions. This project will test the proposition that AI-assisted ‘unbiased’ modelling provides an accurate approach for in silico prediction of hERG liability of small molecules. Global AI-driven docking to hERG of a curated drug database, using multiple AI docking platforms, will be combined with coarse-grained molecular dynamics, informed by extant cryo-EM structures and validated against mutagenesis and electrophysiology data. The integrated datasets that emerge will be used to train an open-access deep learning model that can predict the structure-function binding profile and potency for uncharacterized drugs.

P13: Computational modeling of membrane proteins to study transport mechanisms of novel substrates

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Abstract

Transport proteins play a crucial role in maintaining cellular homeostasis by regulating the movement of small molecules across biological membranes. Found across all kingdoms of life, they are essential for nutrient uptake, ion balance, metabolite exchange, and drug absorption, making them central to understanding both fundamental cell biology and disease-relevant physiology. Additionally, these proteins are key engineering targets when using cell factories in biotechnology for the production of medicines, biochemicals, and materials. Despite the diversity and importance of membrane transporters, many proteins across these groups remain poorly characterized, and their substrate specificities and transport mechanisms are only partially understood. This project investigates the molecular and functional properties of selected Solute Carrier Family (SLC) transporters, Major Facilitator Superfamily (MFS) transporters, and Yeast mitochondrial transporters, to better understand their substrate specificity and transport mechanisms. After having identified novel substrates for selected transporters using laboratory techniques such as untargeted Liquid Chromatography-Mass Spectrometry (LC-MS), we tested different computational methods to explore how molecular interactions within the transporters' binding pockets influence substrate affinity and transport. A range of computational protein-ligand interaction tools was employed to probe substrate binding and transporter functions, including Boltz2, AutoDock Vina, and SPOT. Across all methods tested, results were inconsistent, with control experiments not yielding the anticipated results, suggesting that existing tools may not be well-suited for studying transporters. This work thus serves as a critical evaluation of available tools, and points toward the need for tailored methods or experimental validation to advance our understanding of this important class of proteins.

P14: Decoding RNA-Lipid Interaction in Self-assembly Lipid Nanoparticles through All-Atoms MD microsecond timescale

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Abstract

Although lipid nanoparticles (LNP) have been clinically widely validated for the delivery of RNAs, the use of LNP encapsulating therapeutic RNAs to treat central nervous system (CNS) diseases remains a challenge because of stability issues related to LNP [1]. In this context, lipid self-assembling nanoparticles (SANPs) have been successfully used for the delivery of nucleic acids, showing physical-chemical characteristics with low cytotoxicity and excellent miRNA internalization in glioblastoma cells [2]. Unlike LNPs, SANPs are hybrid lipid-based nanostructures composed by cationic lipids (DOTAP) associated with neutral lipids (Cholesterol) and PEGylated lipid (cer-PEG2000) to encourage the transfection efficiency. Moreover, unlike classical LNPs, they do not exhibit a fully ordered lipid bilayer but a core-shell architecture with the “hydrated corona” including PEGylated lipids, a lipid central shell with cholesterol and DOTAP, and the inner core composed of calcium phosphate (CaP) and encapsulated RNA (Figure 1A). On such basis, we have undertaken microsecond-long atomistic Molecular Dynamics simulations (aa-MDs) to investigate the role of lipid composition, in particular PEGylated lipids cer-PEG2000, in the structural organization of SANPs and in the interaction with a therapeutically designed miRNA and compared them with a classical bilayer system. Our promising results will prompt us to convert our systems to coarse-grained (CG) resolution, to capture larger-scale structural dynamics and longer timescale behaviors of miRNA-SANPs interactions.

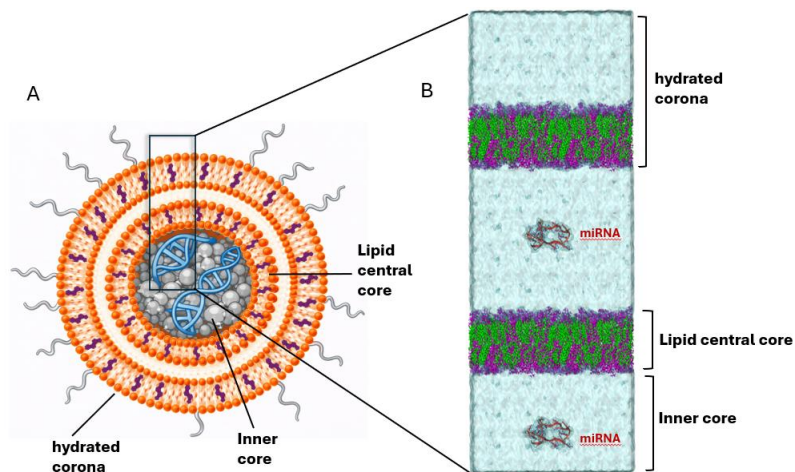


Figure 1: (A) General representation of SANPs architecture and (B) Atomistic reconstruction.

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P15: Coarse-grained molecular dynamics simulations of anti-cancer supramolecular nanotubes

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Abstract

Supramolecular structures have diverse applications, including drug delivery.¹ A well-known drug molecule, camptothecin (CPT), is a proven cancer therapeutic that has been combined with other molecules to increase its biological activity and solubility properties in physiological conditions.² TT1 (a type of tubustecan) is one of such structures and is more effective at treating certain types of cancer in biological studies when compared to CPT alone, owing to its increased efficacy due to the self-assembling nanotubes it forms.³ Studies involving TT1 using Molecular Dynamics (MD) simulations have focused on the initial stages of nanotube formation.⁴ Expanding on these results by simulating a pre-formed nanotube made up of TT1 using MD coarse-grained (CG) simulations using the Martini 3 coarse-grained (CG) simulations and all-atomistic (AA) methods can allow us to understand these structures better by being able to run long-time simulations.⁵

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P16: Evaluating MARTINI 3 Through Influenza M2 Aggregation in Lipid Bilayers

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Abstract

The Influenza A virus M2 protein is a membrane-embedded proton channel essential for viral replication and budding. Beyond its role in proton transport, M2 is known to cluster within lipid bilayers and contribute to the membrane curvature, a key step in viral release. Previous studies, including our work using the Martini 2 force field, have shown that M2 aggregation is influenced by lipid composition, particularly cholesterol. However, the extent to which these behaviors are captured by the Martini 3 force field remains unclear. In this study, we used coarse-grained molecular dynamics (CGMD) simulations to investigate M2 aggregation across different lipid bilayers and assess the effects of Martini 3 on protein clustering and membrane behaviour. The effect on aggregation of amphipathic helices (AHs) was explored.

References

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CONFERENCE PROGRAM

Day 1 — Martini Fundamentals

Time		Activity
09:30 10:00	–	Registration
10:00 10:10	–	Welcome — E. Mikros, A. Kolocouris
10:10 10:55	–	Introduction to Coarse-Graining and Martini — Paulo C. T. Souza
10:55 11:40	–	Martini models for biomolecules / Martini ecosystem — Luís Borges Araújo
11:40 12:10	–	Coffee break
12:10 12:55	–	Advanced tools / TS2CG — Ilias Patmanidis & Fabian Schuhmann
13:00 14:00	–	Lunch
14:00 17:00	–	Hands-on Session 1: Basics of Martini (membranes, proteins, martinize2, TS2CG)
17:00 19:00	–	Poster session & gathering

Day 2 — Pharmaceutical Applications

Time		Activity
09:30 10:10	–	Transport to the Athens Life Sciences Technology Park of ELPEN by bus from NKUA.
10:20 11:00	–	Tour in the labs of Life Sciences Technology Park of ELPEN b
11:00 11:10	–	Welcome — ELPEN representative
11:10 11:55	–	Parametrization of Martini models — Paulo C. T. Souza

Time		Activity
11:55 12:40	–	Modelling protein-ligand interactions using Martini — Matthieu Chavent & Magdalena Szczuka
12:40 12:50	–	Coffee break
12:50 13:35	–	Computational modelling of liposomes / LNPs — Paulo C. T. Souza & Mariana Valério
13:35 14:20	–	Lunch in ELPEN
14:20 15:00	–	Transport back from ELPEN to NKUA by bus
15:00 18:00	–	Hands-on Session 2: Parametrization, ligand binding, liposomes/LNPs
19:30		Speaker dinner

Day 3 — Biological Applications

Time		Activity
10:00 10:10	–	Welcome — Robin Corey
10:10 10:55	–	Transmembrane proteins — Dimitrios Kolokouris & Robin Corey
10:55 11:40	–	Peripheral membrane proteins — Saman Fatihi & Nishita Mandal
11:40 12:10	–	Coffee break
12:10 12:55	–	Complex biological assemblies with bentopy — Chelsea Brown & Marieke Westendorp
13:00 14:00	–	Lunch
14:00 17:00	–	Hands-on Session 3: Building, running, analysing and visualising membrane protein simulations

Time	Activity
17:00 – 20:00	Optional Acropolis/museum visit

Organisers

- **École normale supérieure de Lyon** — Paulo C. Telles de Souza
- **University of Bristol** — Robin Corey
- **National & Kapodistrian University of Athens** — Antonios Kolocouris

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